

HMSR: Multi-Cohort Sarcopenia Transcriptomic Resource

WP2: 4 Cohorts | 135 samples | 53 sarcopenia cases | 82 controls

HSS
(GSE111006)
n=32

JSS
(GSE111010)
n=23

SSS
(GSE111016)
n=40

MOS
(GSE226151)
n=40

WP3: OLS DE per cohort + DerSimonian-Laird random-effects meta
37,469 genes · BH-FDR genome-wide

WP3 Gate: meta-p < 0.005 · direction = 1.0 · LOCO = 1.0 · I² < 50%
→ 195 candidate genes (141↓ 54↑)

Candidate set enriched for mitochondrial genes (26/195, ~9.5x)
Highlighted: 6 mitochondrial (2 OXPHOS) + CLTB (non-mito outlier)

WP5: Exercise epigenetic evidence ladder
37 exercise epigenetic rows | 90 human multilayer rows